

Experimental Networked Embedded Mini Drone – Part II. Distributed FDI

A. Tanwani, S. Gentil, S. Lesecq, J-M. Thiriet

Abstract— Embedded systems constitute a category whose safety is critical and where FDI real time constraints are particularly important. Embedded algorithms must be the simplest as possible and computations may be distributed between the embedded system and a more powerful distant computer. This paper proposes a bank of observers to diagnose faults of a small helicopter controlled in closed loop. The studied prototype is a 4 rotors mini-drone equipped with an attitude central for system positioning, made of 3 rate gyros, 3 accelerometers and 3 magnetometers. A dedicated network architecture is proposed.

Keywords— Fault Diagnosis, Embedded System, Generalized Observer Scheme Residual Generation, Distributed Computer Control System, Network Topology, Drones.

I. INTRODUCTION

DRONES constitute a category of embedded systems whose safety is critical. The system must still be able to exhibit given properties, such as stability or precision, even in the presence of faults. Compared to aircrafts, they constitute low cost systems whose sensors and actuators are often subject to faults (offsets, drifts). Consequently, their maintenance is essential and sometimes, difficult. Hardware redundancy is not conceivable, not only due to cost but also to constraints on weight, energy and operating space.

Real time constraints are particularly important for these systems. For this reason, early fault detection is necessary but embedded algorithms must be as simple as possible. Thus computations may be distributed between the embedded system and a more powerful distant computer sharing information through a network. Control, state estimation, fault detection, fault isolation and controller accommodation or reconfiguration must be considered as cooperative functions distributed onto the network architecture. Splitting of algorithms is an open problem.

NCS emerged recently with the rapid development of wire or wireless network technology. It is thus an important topic to take into account the network for the design of the

control and FDI algorithms [1], [2]. Evidently, from the FDI point of view the design of residuals and their evaluation is more difficult [3]. Some faults are induced by the network, such as packet losses which are comparable to an intermittent sensor failure. Delays are also induced by the network and should be taken into account by the algorithms [4].

This paper deals with a particular application. This preliminary work is completed in simulation. Section II presents the studied prototype: a 4 rotors mini-drone equipped with an attitude central for system positioning, made of 3 rate gyros, 3 accelerometers and 3 magnetometers. A model of the system has been designed for the control synthesis, simulated under the Matlab-Simulink platform and further extended to fault analysis. Several kinds of realistic faults are hypothesized on this system: their study is proposed in [5]. This paper focuses on the case of sensor and actuator faults. Section III summarizes briefly the closed loop control through a state observer. In section IV, a model-based diagnostic approach is adopted. An FDI algorithm based on a generalized observer scheme is proposed and results are commented. Section V proposes a network architecture to embed and remotely distribute control and FDI algorithms.

II. PRESENTATION OF THE SYSTEM

A drone with 4 rotors is regarded as a composition of two PVTOL (Planar Vertical Take off Landing) problems whose axes are orthogonal allowing a movement of six degrees of freedom.

A. Mechanical model

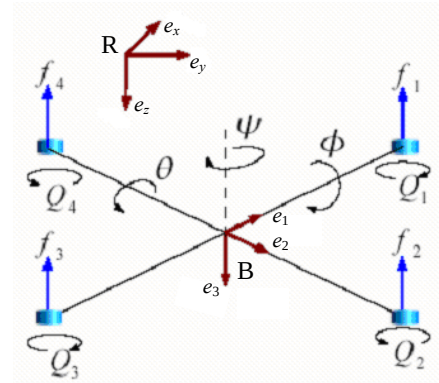


Fig. 1. Coordinate frames assigned to the drone

Each one of the four electric motors, driving the four

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S. Gentil, S. Lesecq, J-M. Thiriet are with the Grenoble Images Parole Signal Automatique Lab (GIPSA-Lab UMR 5216 CNRS, INPG, UJF), Control Systems Dept., BP 46, 38402 St Martin d'Hères France. Phone: (33) 476-82-62-39; fax: (33) 476-82-63-88; {Suzanne.Lesecq, Sylviane.Gentil, Jean-Marc.Thiriet}@inpg.fr.

A. K. Tanwani was with the Laboratoire d'Automatique de Grenoble (UMR 5528 CNRS, INPG, UJF). He is now with the University of Illinois at Urbana Champaign (UIUC).

blades, provides the thrust f_i for the movement of the drone [6], Fig. 1. The speed and the moment of inertia of rotor i are indicated, respectively, by ω_i and I_r . τ_i represents the electromotive torque of rotor i . The mechanical equation for each motor is expressed as:

$$I_r \dot{\omega}_i = \tau_i - Q_i, i \in \{1, 2, 3, 4\} \quad (1)$$

The reactive torque generated by rotor i in (1) is given by:

$$Q_i = k \omega_i^2 \quad (2)$$

Thus, the total thrust generated by the four rotors is:

$$T = \sum_{i=1}^4 f_i = b \sum_{i=1}^4 \omega_i^2 \quad (3)$$

$k > 0$ and $b > 0$ are two parameters depending on the air density, the radius, the shape, the pitch angle of the blade and other factors.

To develop the dynamic equations of the drone, consider R, an inertial frame of reference (e_x, e_y, e_z in Fig. 1) and B, a frame attached to the drone (e_1, e_2, e_3 in Fig.1).

The model is described by the following equations:

$$\dot{p} = v, \quad (4)$$

$$\dot{v} = g e_z - \frac{1}{m} C^T(q) T \quad (5)$$

$$\dot{q} = \frac{1}{2} q \otimes \Omega = \frac{1}{2} \Omega(\varpi) q \quad (6)$$

$$I_f \dot{\varpi} = -\varpi \times I_f \varpi - G_a + \tau_a \quad (7)$$

m indicates the mass of the drone. g is the acceleration due to the gravity. $e_z = (0, 0, 1)^T$ is a unit vector in reference frame R. The vector $p = (x, y, z)^T$ indicates the position of the origin of frame B relative to frame R. The vector $v = (v_x, v_y, v_z)^T$ is the linear velocity of the origin of B measured in the reference frame R. $I_f \in \mathbb{R}^{3 \times 3}$ is a symmetric positive definite constant inertia matrix of the drone with respect to frame B whose origin is at the centre of mass and ϖ is the angular velocity of the drone measured by the rate gyros in the B frame.

The gyroscopic torques in (7), due to the combination of the rotation of the drone and the four rotors, are modeled as:

$$G_a = \sum_{i=1}^4 I_r (\varpi \times e_z) (-1)^{i+1} \omega_i \quad (8)$$

The torques generated by the rotors in (7) are given by:

$$\begin{aligned} \tau_a^1 &= db(\omega_2^2 - \omega_4^2) \\ \tau_a^2 &= db(\omega_1^2 - \omega_3^2) \\ \tau_a^3 &= k(\omega_1^2 + \omega_3^2 - \omega_2^2 - \omega_4^2) \end{aligned} \quad (9)$$

where d is the distance from the rotors to the centre of mass of the drone.

In (6), the attitude dynamics are represented by $q = (q_0, \vec{q}^T)^T$ which is a rotational quaternion and $\Omega = (0, \varpi)^T$ is also a quaternion [7], [8]. A unit quaternion represents the axis of rotation $\vec{k}$ and the angle of rotation

ζ around this axis under the form $q = \begin{bmatrix} \cos \frac{\zeta}{2} & \vec{k}^T \sin \frac{\zeta}{2} \end{bmatrix}^T$.

In (6) q corresponds to the change of coordinates between the two frames R and B that gives the orientation of the drone. If $\vec{r}$ is a vector expressed in R, then its coordinates in B are given by:

$$b = q^{-1} \otimes r \otimes q \quad (10)$$

where $r = (0, \vec{r}^T)^T$ and $b = (0, \vec{b}^T)^T$ and q^{-1} is the complementary rotation of the quaternion q defined by

$$q^{-1} = [q_0, -\vec{q}^T]^T = \bar{q} \quad (11)$$

$\otimes$ indicates the quaternion multiplication. (10) can also be expressed as:

$$\vec{b} = C(q) \vec{r} \quad (12)$$

where $C(q)$ (used in (5)) is the Rodrigues matrix defined as:

$$C(q) = (q_0^2 - \vec{q}^T \vec{q}) I + 2(\vec{q} \vec{q}^T - q_0 [\varpi^\times]) \quad (13)$$

This leads to the following definition of $\Omega(\varpi)$ in (6):

$$\Omega(\varpi) = \begin{pmatrix} 0 & -\varpi^T \\ \varpi & -[\varpi^\times] \end{pmatrix} \quad (14)$$

where $[\varpi^\times]$ is the self cross product:

$$[\varpi^\times] = \begin{pmatrix} 0 & -\varpi_3 & \varpi_2 \\ \varpi_3 & 0 & -\varpi_1 \\ -\varpi_2 & \varpi_1 & 0 \end{pmatrix} \quad (15)$$

B. Sensors

The estimation of attitude (or orientation), position and speed is a pre-requisite for flight control. To achieve the prerequisite, a central navigation unit is used that consists of a tri-axes accelerometer (a_1, a_2, a_3), three magnetometers (m_1, m_2, m_3) and three rate gyros (g_1, g_2, g_3) mounted at right angle. These sensors use MEMS (Micro Electronic and Mechanical Systems) technology to reduce the weight, dimension and cost.

1) Rate Gyroscopes: the angular velocity ϖ is measured by three rate gyros. This kind of sensor is usually affected by various errors, such as fixed bias, drift bias, fixed scale or asymmetric scale factor error, and noise. Theoretically, the integral of ϖ gives the relative orientation but the presence of bias at low frequencies generates errors which accumulate over time. The sensor measurements are modeled as:

$$\varpi_g = \varpi + \beta + \eta_1, \quad \dot{\beta} = -\frac{1}{\tau} I_3 \beta + \eta_2 \quad (16)$$

where ϖ_g is the sensor value, β is the bias, $\tau = 100$ s is the time constant obtained from an Allan Variance chart, and η_i , $i = 1, 2$ are assumed to be Gaussian white noises.

2) Accelerometers: The 3-axes accelerometer senses inertial forces and gravity in the aircraft body coordinate system B. The transformation of force and gravity from

inertial frame R to body frame B is given by:

$$b_{acc} = C(q)(a - ge_z) + \eta_{acc} \quad (17)$$

where b_{acc} corresponds to the three values measured by the sensors. The motion is supposed quasi static so that linear accelerations a are neglected. In this way, accelerometers are only sensible to the earth gravitational pull.

3) Magnetometers: the information provided by the three magnetometers mounted orthogonally is added to the inertial measurements. The magnetic field vector is transformed to the mobile frame attached to the aircraft and is defined by:

$$b_{mag} = C(q)h_m + \eta_{mag} \quad (18)$$

where $h_m = (h_{mx} \ 0 \ h_{mz})^T$ and b_{mag} are the three components of the magnetic field vector in R and B, respectively.

It can be observed that the accelerometers and magnetometers measurements are modeled by static non linear equations measuring constant vectors ge_z and h_m with a matrix depending on the drone attitude expressed by q .

III. STATE OBSERVER AND CLOSED LOOP CONTROL

The measurements provided by the above mentioned accelerometers, magnetometers and rate gyroscopes are used to feed an observer. The speed and quaternion estimation are used in a feedback loop. The reference corresponds to a quaternion (Fig. 2). For simplification, $q = (1 \ 0 \ 0 \ 0)^T$ is chosen as reference for the following simulations. The aim of the present work being the evaluation of the diagnostic task for the 4-rotors helicopter when a network is embedded, no comparison of different control laws is performed. Among various possibilities (for instance, see [6], [8]), the controller implemented for the attitude stabilization is similar to the one detailed in [6].

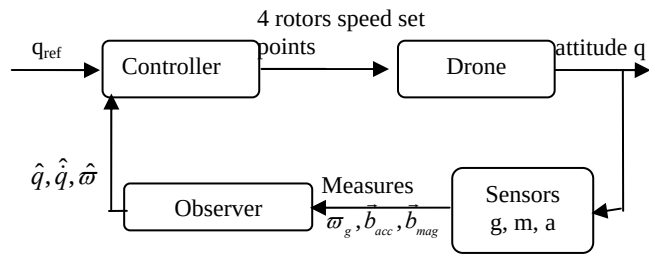


Fig. 2. The 4-rotor helicopter control loop

Several state observers have been implemented [9]. The following observer is inspired by Extended Kalman Filtering technique. The quaternion model (6) is considered as the dynamic state equation, to be used for state prediction. The state matrix in (6), $\Omega(\varpi)$, depends on the values of the drone angular velocity. The measurements ϖ_g provided by the rate gyros could be used to compute this matrix. But, as shown by (16), these measurements are biased. Thus the model is extended to take into account both the quaternion dynamics and the gyroscope bias. The extended state vector

is:

$$X^T = [q_0 \ q_1 \ q_2 \ q_3 \ \beta_1 \ \beta_2 \ \beta_3] \quad (19)$$

(6), (14), (15), (16) show that the first 4 components of X are given by a non linear relation

$$\dot{q}(t) = f(q(t), \varpi(t), \beta(t)) \quad (20)$$

while $\dot{\beta}(t)$ is given by the linear relation (16). (20) is linearized, which results in matrix A :

$$A = \frac{\partial f}{\partial X} = \begin{bmatrix} 0 & -(\varpi_1 - \beta_1) & -(\varpi_2 - \beta_2) & -(\varpi_3 - \beta_3) & q_1 & q_2 & q_3 \\ -(\varpi_1 - \beta_1) & 0 & -(\varpi_3 - \beta_3) & -(\varpi_2 - \beta_2) & -q_0 & q_3 & -q_2 \\ -(\varpi_2 - \beta_2) & -(\varpi_3 - \beta_3) & 0 & -(\varpi_1 - \beta_1) & -q_3 & -q_0 & q_1 \\ -(\varpi_3 - \beta_3) & -(\varpi_2 - \beta_2) & -(\varpi_1 - \beta_1) & 0 & q_2 & -q_1 & -q_0 \\ 0 & 0 & 0 & 0 & -\frac{1}{\tau} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{\tau} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{\tau} \end{bmatrix} \quad (21)$$

and matrix B :

$$B = [0_4 \ I_3]^T \quad (22)$$

$$\dot{X} = AX + B\eta_2 \quad (23)$$

Then the linearized model (23) is discretized and considered in the Extended Kalman Filter to obtain the prediction part of the filter $\hat{X}_{k+1/k}$ whose first 4 components give the quaternion prediction $\hat{q}_{k+1/k}$.

The output relation is constituted by (17) and (18). It can be rewritten as:

$$\begin{bmatrix} b_{acc} \\ b_{mag} \end{bmatrix} = h(q) + \begin{bmatrix} \eta_{acc} \\ \eta_{mag} \end{bmatrix} \quad (24)$$

The idea is to compute $\hat{q}_{ms}$ considering that (24) is a non linear system of 6 equations in q to be solved. In the literature, $\hat{q}_{ms}$ is called the pseudo-measurement of the quaternion.

$$\hat{q}_{ms} = \arg \min_{\|q\|_2=1} \left\{ \left(\begin{bmatrix} b_{acc} \\ b_{mag} \end{bmatrix} - h(q) \right)^T \left(\begin{bmatrix} b_{acc} \\ b_{mag} \end{bmatrix} - h(q) \right) \right\} \quad (25)$$

The criterion to be minimized can be expressed as:

$$J(q) = \sum_{j=1}^6 (b_{mod}(j) - b_{mes}(j))^2 \quad (26)$$

where $b_{mod} = C(q)\vec{r}$, $\vec{r} = [0 \ 0 \ g \ h_m^T]^T \in \mathbb{R}^{6 \times 1}$ and

$$b_{mes} = \begin{bmatrix} b_{acc}^T & b_{mag}^T \end{bmatrix}^T$$

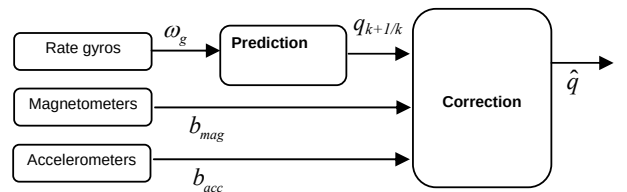


Fig. 3. The observer principle for the control synthesis

The prediction $\hat{q}_{k+1/k}$ is corrected by the pseudo-measurement (Fig. 3):

$$\hat{q} = q_{k+1/k} + K(q_{ms} - q_{k+1/k}) \quad (27)$$

The second part of (27) is a quaternion difference. Its evaluation is explained in the next section.

IV. FDI ALGORITHM

As a state observer has been designed for the control synthesis, it seems quite natural to investigate if this observer can be adapted to the diagnostic purpose. The use of a bank of observers to design structured residuals is well known [10] and has been applied recently to an aircraft model [11].

A. Diagnosis of Sensors

To isolate a fault concerning one of the output sensors, under the hypothesis that actuators are fault-free, a Generalized Observer Scheme (GOS) is used. Based on this principle, an observer that takes 8 out of 9 system outputs is used to generate structured residuals. The basic idea is to estimate the quaternion using these 8 sensor values and compare it to the quaternion calculated via the drone mechanical model (Fig. 4). The residual is thus insensitive to the faults in the sensor that is not used and the signature table is strongly isolable.

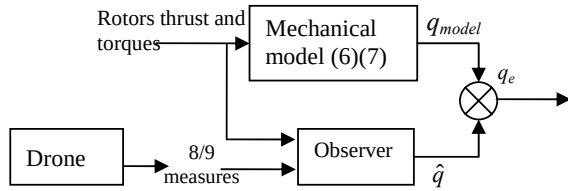


Fig. 4. Residuals generation using observers

The quaternion of error q_e measures the discrepancy between the quaternion obtained with one of the observer within the observer bank $\hat{q}$ and the quaternion computed with the mechanical equation of the drone q_{model} . It is given by:

$$q_e = \hat{q} \otimes q_{model}^{-1} \quad (28)$$

Since this is a quaternion of rotation, it can be expressed in the following form:

$$q_e = \begin{bmatrix} q_{e0} & \overline{q_e}^T \end{bmatrix}^T = \begin{bmatrix} \cos\left(\frac{\phi_e}{2}\right) & u_e^T \sin\left(\frac{\phi_e}{2}\right) \end{bmatrix}^T \quad (29)$$

ϕ_e is chosen as a residual. Actually, it has some physical meaning: it is the difference between the original orientation and the estimated orientation of the drone. It is computed in degrees:

$$\phi_e = 2 \frac{180}{\pi} \arccos(q_{e0}) \quad (30)$$

The observer that estimates the quaternion using 3 accelerometers, 3 magnetometers and 3 rate gyros has

already been explained in the previous section. Thus, the problem that must be addressed is the quaternion estimation while neglecting one of the sensor values.

When the FDI algorithm discards a gyroscope measurement, there is a difficulty to apply the linearized model (21) because it explicitly uses the 3 measurements ω_i to compute matrix A . The measurement ω_i that is discarded is in fact replaced in A by the variable $\hat{\omega}_i$ which represents the computation of this component of the drone angular velocity by model (7).

Table I is the fault signature table obtained in case of faults in the three rate gyros. It shows that the estimator *gyro1* is sensitive to faults in all the sensors ($f_{a1}, f_{m1}, f_{g1}, i = 1, 2, 3$) except f_{g1} , which indicates a fault in the rate gyroscope 1 (along the x-axis). The same definition applies to all other symbols. Thus, it is possible to isolate the faults in rate gyros by looking at the residuals generated by the three observers mentioned in the first column of Table I.

TABLE I
FAULT SIGNATURE FOR DIAGNOSIS OF RATE GYROS

	f_{a1}	f_{a2}	f_{a3}	f_{m1}	f_{m2}	f_{m3}	f_{g1}	f_{g2}	f_{g3}
<i>gyro1</i>	1	1	1	1	1	1	0	1	1
<i>gyro2</i>	1	1	1	1	1	1	1	0	1
<i>gyro3</i>	1	1	1	1	1	1	1	1	0

The same idea applies to the surveillance of accelerometers and magnetometers. The optimization algorithm (25) generates an optimal quaternion using five out of six measurements of accelerometers and magnetometers.

TABLE II
FAULT SIGNATURE FOR DIAGNOSIS OF ACCELEROMETERS AND MAGNETOMETERS

	f_{a1}	f_{a2}	f_{a3}	f_{m1}	f_{m2}	f_{m3}	f_{g1}	f_{g2}	f_{g3}
<i>acc1</i>	0	1	1	1	1	1	0	0	0
<i>acc2</i>	1	0	1	1	1	1	0	0	0
<i>acc3</i>	1	1	0	1	1	1	0	0	0
<i>mag1</i>	1	1	1	0	1	1	0	0	0
<i>mag2</i>	1	1	1	1	0	1	0	0	0
<i>mag3</i>	1	1	1	1	1	0	0	0	0

Table II shows that the estimator *acc1* is insensitive to faults in the accelerometer along the x-axis (f_{a1}). The '1's in the first row show that a non-zero residual is generated by *acc1* if there appears a fault in any other accelerometer or magnetometer. Note that all these six estimators are insensitive to faults in rate gyros.

B. Diagnosis of Actuators

The 4 actuators are independent. The model of each electrical motor has been used for their surveillance. Due to the constraint imposed by the availability of a single sensor that measures the rotor speed, the principle shown in Fig. 5

is used for fault diagnosis of each rotor. Hence, the fault signatures are given in Table III. The actuator fault localization cannot be performed with this technique. For instance, a fault in the rotational speed sensor or in the corresponding motor cannot be differentiated. This limitation could be removed by additional sensors to measure other variables such as current or voltage but this solution is contrary to the weight constraint.

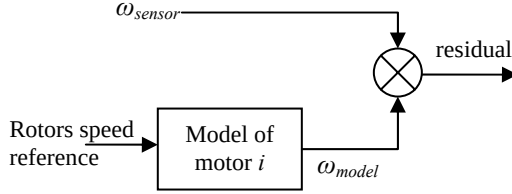


Fig. 5. Residual generation for actuators

TABLE III
FAULT SIGNATURE FOR DIAGNOSIS OF ACTUATORS

	$f_{rotor 1}$	$f_{rotor 2}$	$f_{rotor 3}$	$f_{rotor 4}$
<i>Rotor1</i>	1	0	0	0
<i>Rotor2</i>	0	1	0	0
<i>Rotor3</i>	0	0	1	0
<i>Rotor4</i>	0	0	0	1

C. Simulation results

The closed loop system in Fig. 2 is simulated with Matlab Simulink and various faults have been injected into the system. The FDI performance is now discussed.

In the first experiment (Fig. 6), the information delivered by the rate gyro g_2 along the y-axis is assumed to be lost at time $t = 5s$ (breakdown of sensor g_2 , fault f_{g_2} in Tables I and II). The residuals generated by $acc1$, $acc2$, $acc3$ (first row in Fig. 6) and $mag1$, $mag2$, $mag3$ (second row) remain null even after the fault has occurred. This is consistent with the design of estimators $acc1$ to $mag3$ that do not depend on the gyro measurements. Residuals generated by $gyro1$ and $gyro3$ are non zero after $t = 5$ and $gyro2$ is insensitive to f_{g_2} (third row), which is consistent with Table II.

In the second experiment, a failure of the magnetometer m_1 along the x-axis is considered. The fault f_{m1} (breakdown of m_1) is simulated at $t = 4s$. In Fig. 7, the residuals generated by the six estimators defined in Table II are shown. The estimator $mag2$ is clearly insensitive to f_{m1} while the other five estimators exhibit a high value after the fault has been introduced. Thus, the results are consistent with the theoretical formulation.

Note that various fault scenarios have been simulated for the sensor faults. For instance, faults such as the decalibration of a sensor gain, or the introduction of an offset drift on the sensor value have been studied [9]. They are not given in this paper, due to the constraint on the paper length.

In Fig. 8, a wiring fault in rotor 1 is considered at time $t = 6s$ (wiring disconnection). The fault is simulated at $t = 5s$ and the residual increases immediately after the fault has

occurred.

The observer bank scheme (Fig. 4) uses the drone dynamical model (7) to compute the reference quaternion q_{model} with Eq. (6). Then the different residuals are computed with (30). As a consequence, ϕ_e is sensitive to the parameters in (7), especially to the inertia matrix I_f .

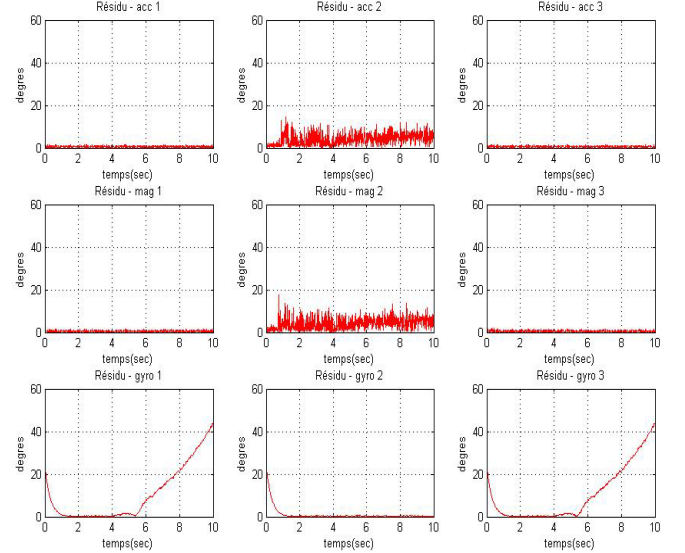


Fig. 6. Residuals generated by the 9 observers or estimators in the case of the breakdown in the rate gyro along the y-axis (f_{g_2}).

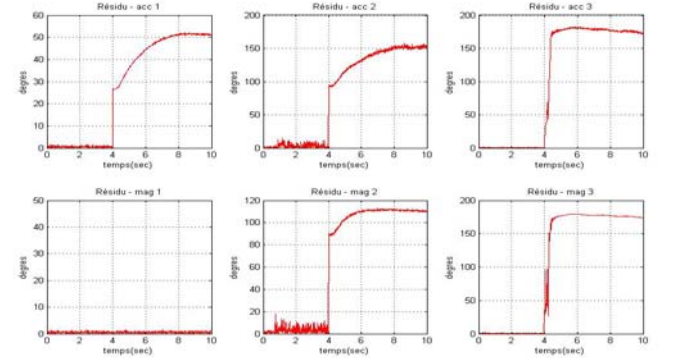


Fig. 7. Residuals generated by the 6 estimators (Table II) in the case of the breakdown in the magnetometer along the x-axis (f_{m1}).

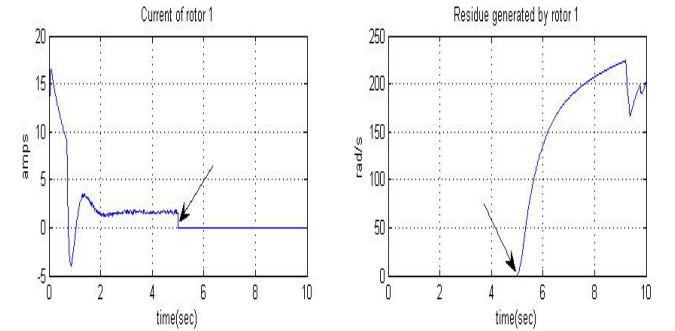


Fig. 8. Wiring fault in rotor 1.

Fig. 9 shows the influence of a 10% error in the identification of the inertia matrix. As expected, due to the integration step, this error induces a slow divergence of the residuals. However, when a fault occurs, the residual are still

sensitive (Fig. 10, breakdown of $m1$ at time $t = 6s$), and the results remain consistent with Tables I and II.

Note that an inappropriate estimation of parameter b in (9) will induce the same consequences. The filter bank that will be implemented in the benchmark must be insensitive to some model parameter errors. Thus, adaptive thresholds will be used instead of constant ones.

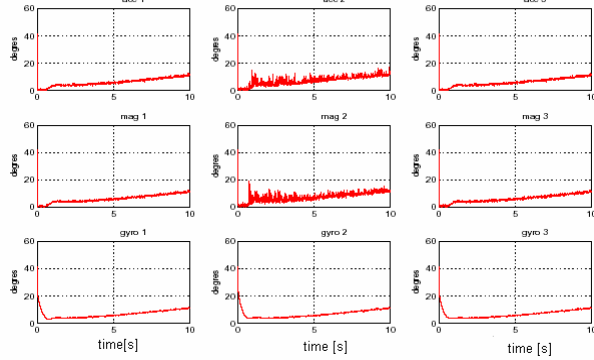


Fig. 9. Influence on the residuals of a 10% error in the identification of I_f

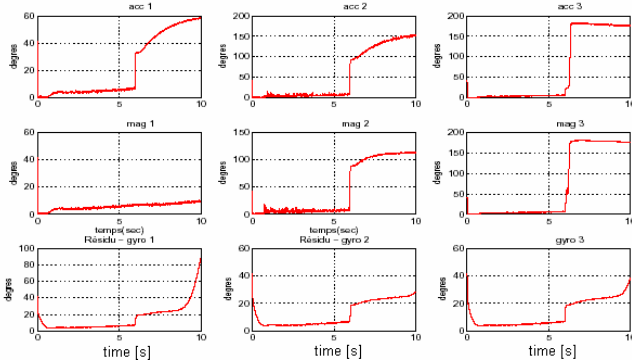


Fig. 10. Residuals in the case of the breakdown of $m1$ when a 10% error in the identification of I_f exists

V. NETWORK ARCHITECTURE

The benchmark presently under development will be equipped with two networks (Fig.11). The first network is an embedded network on the drone-helicopter itself. The control loop is closed through the embedded network that must be designed to take into account the fast dynamics and strong real-time constraints of the system. The problems induced by the network are studied in [5].

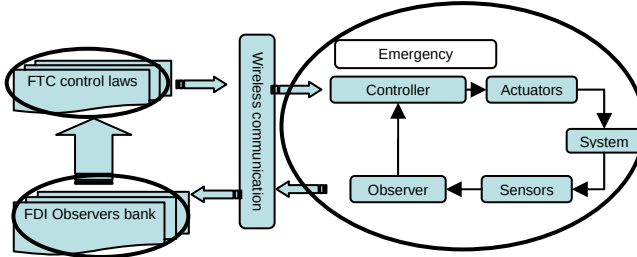


Fig. 11. The two networks

The second network is a wireless network which aims at ensuring the communication between the drone-helicopter (a mobile vehicle) and the ground. It is evident that

computations related to observer bank will be time-consuming. It is thus not reasonable to implement the filter bank on the embedded microcontroller. Thus, the FDI and FTC algorithms will be deported in the ground station. Evidently, the duration of the transmission will be longer, and the real-time constraints will not be met as easily as in the case of the wired network, so some strategies of Networked Fault-Tolerant Control and decentralized Diagnosis will be tested.

VI. CONCLUSION

This paper has presented the FDI of an embedded system with state estimation and closed loop control. The observer which is implemented for the control has been modified to allow fault detection and isolation through a Generalized Observer Scheme. A strongly isolable signature table has been obtained. The sensitivity of the observer bank to model precision has been studied. Different fault scenarios have been presented. The FDI algorithms are time consuming and will be deported on a ground station connected to the drone by a wireless network. The embedded network will only be in charge of the closed loop control. In future work, specific problems will be tested, for instance network induced delays, packet losses due to the worse Quality of Service between the drone and the ground station and the management of the roaming (between several ground stations).

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